Predictive Modeling in Healthcare

Performance Measures

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Motivation

- During and after the derivation of a prediction model, we need to know that its performance is good
- But what is “good performance”? Three domains:

1. Intrinsic Performance
2. Extrinsic (causal) performance
3. Algorithmic Fairness

- I’ll dedicate some slides to #2 in the end, and #3 gets a whole lecture. Today we focus on #1

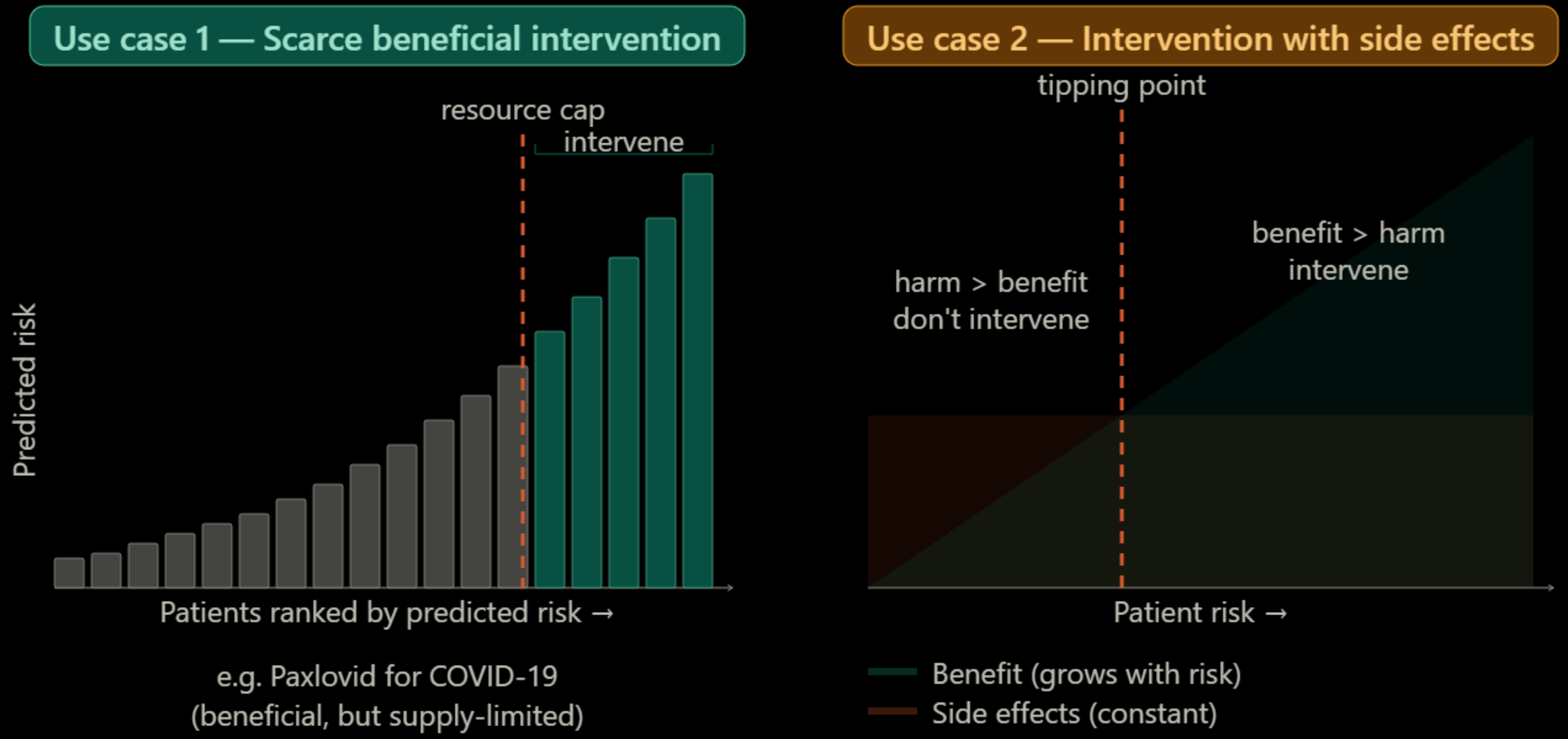
Intrinsic Performance

- Does the model succeed in the specific task it was trained to do (e.g., predicting who gets a heart attack)
- The performance we care about is on previously unseen data (“test set”)
- Our intended use of the model will affect which specific measures we care about more

Use Cases

- We usually have in mind one of two use cases:
- First, we might have some intervention that we consider to be beneficial to everyone (or at least not harmful to anyone), but our resources only permit intervening on a selected population. We then opt to choose those at maximum risk
  - Think Paxlovid for Covid-19
- Second, we might have some intervention (maybe a drug) that has some benefit but also some side effects, and we want to find "the tipping point" of risk, above which it would make sense to intervene
  - Think statins to prevent cardiovascular disease

Use Cases (2)


Use case 1 — Scarce beneficial intervention
resource cap
intervene
Predicted risk
Patients ranked by predicted risk →
e.g. Paxlovid for COVID-19
(beneficial, but supply-limited)
Use case 2 — Intervention with side effects
tipping point
harm > benefit
don't intervene
benefit > harm
intervene
Patient risk →
Benefit (grows with risk)
Side effects (constant)

An Aside – Why Top Risk?

- In medicine (and maybe everywhere), when we have limited resources, we usually default to choosing the patients most at risk
- We have a limited supply of Covid-19 drugs (Paxlovid, etc.), so we give them to those most at risk to experience severe Covid-19
- But this doesn’t necessarily make sense
- Shouldn’t we give them to those that would benefit most? Shouldn’t we strive to improve some societal measure of benefit?
- These are open questions, but understand that the answer is not simple

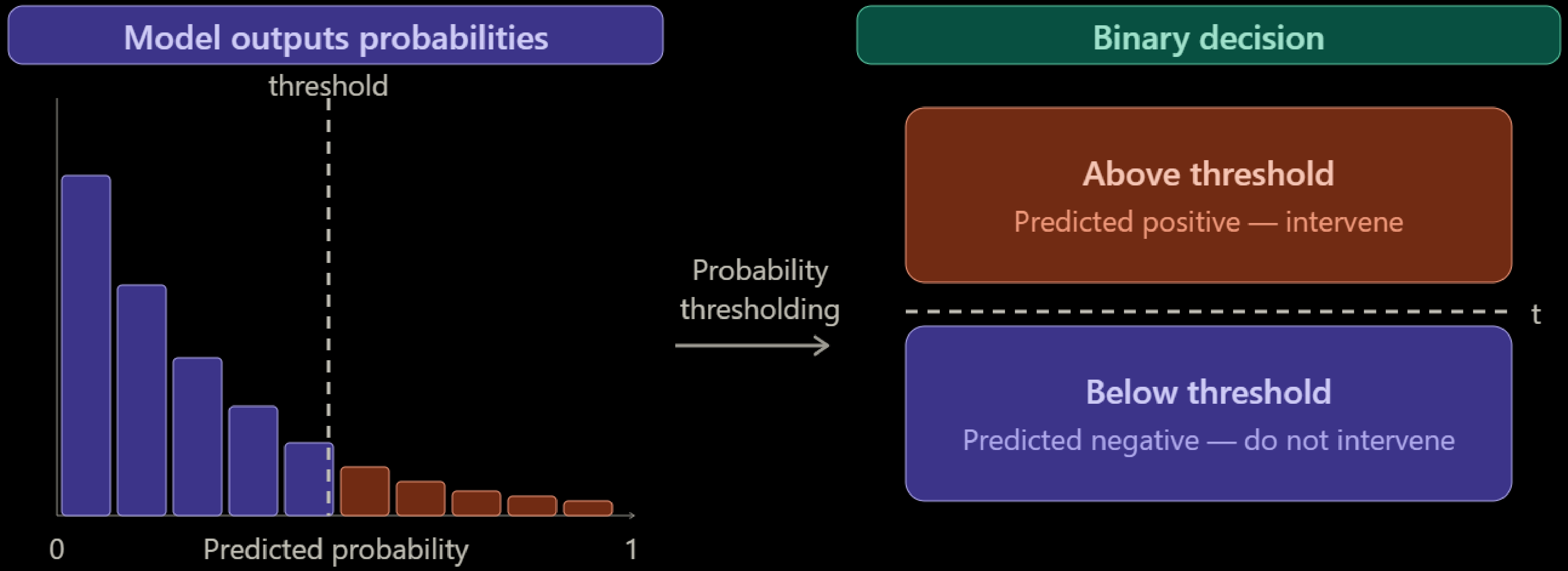
A word on Nomenclature

- Because of the different traditions involved, the same things tend to have different names
  - Sensitivity => Recall
  - Positive Predictive Value => Precision
- I don’t promise to mention each variation

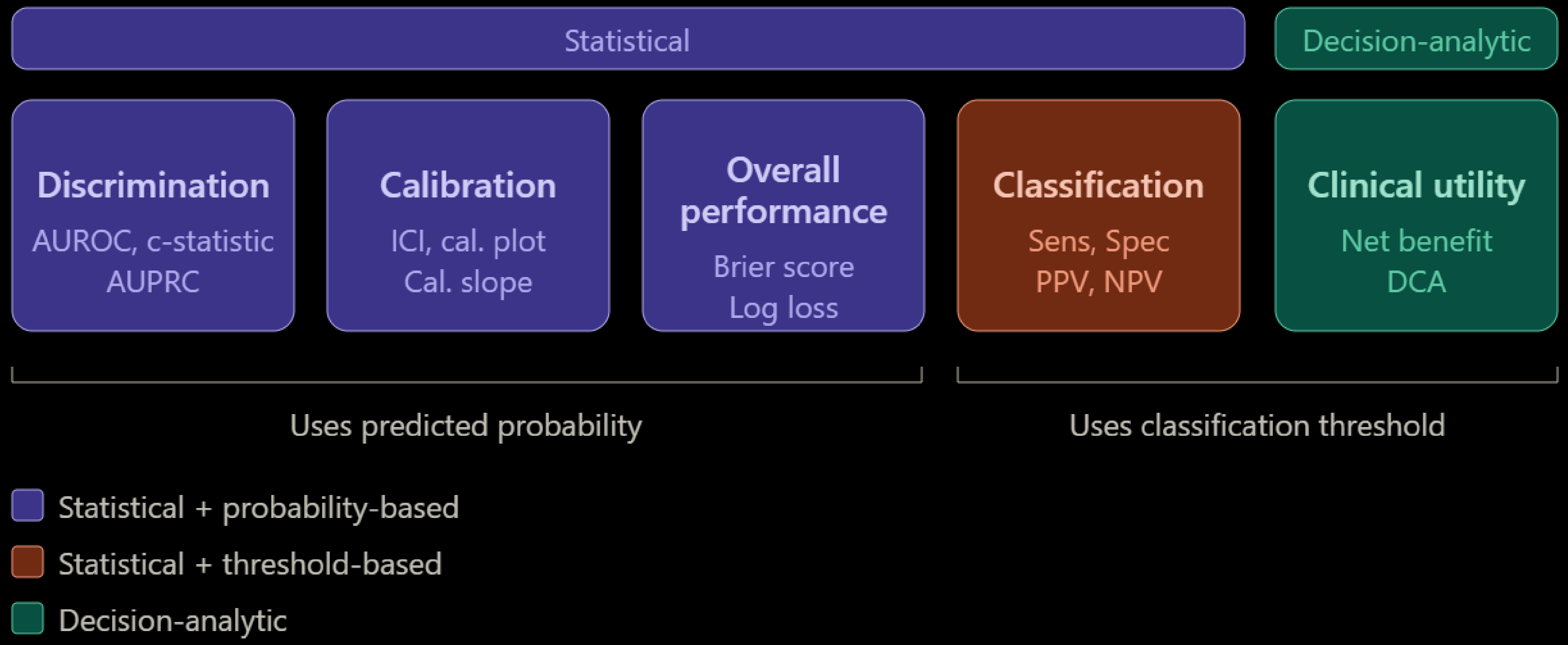
Our Model

- For each person we see $Y \in \{0,1\}$
- We assume that this Y was sampled from a Bernoulli distribution with probability P: $Y \sim \text{Bernoulli}(P)$
- P is a function of the different predictors of the individual, some observed and some not, $P = f(X)$
- We predict $\hat{P}$, and then go to $\hat{Y}$ to make a clinical decision (treat / don't treat) by choosing a threshold

Two-stage Predictions



5 Families of Performance Measures



Characteristics of a Good Metric

- Before we starting going through them, let’s first discuss what characteristics we want from a good metric
- We identify three such characteristics:
  1. Properness
  2. Clear Focus
  3. Interpretability

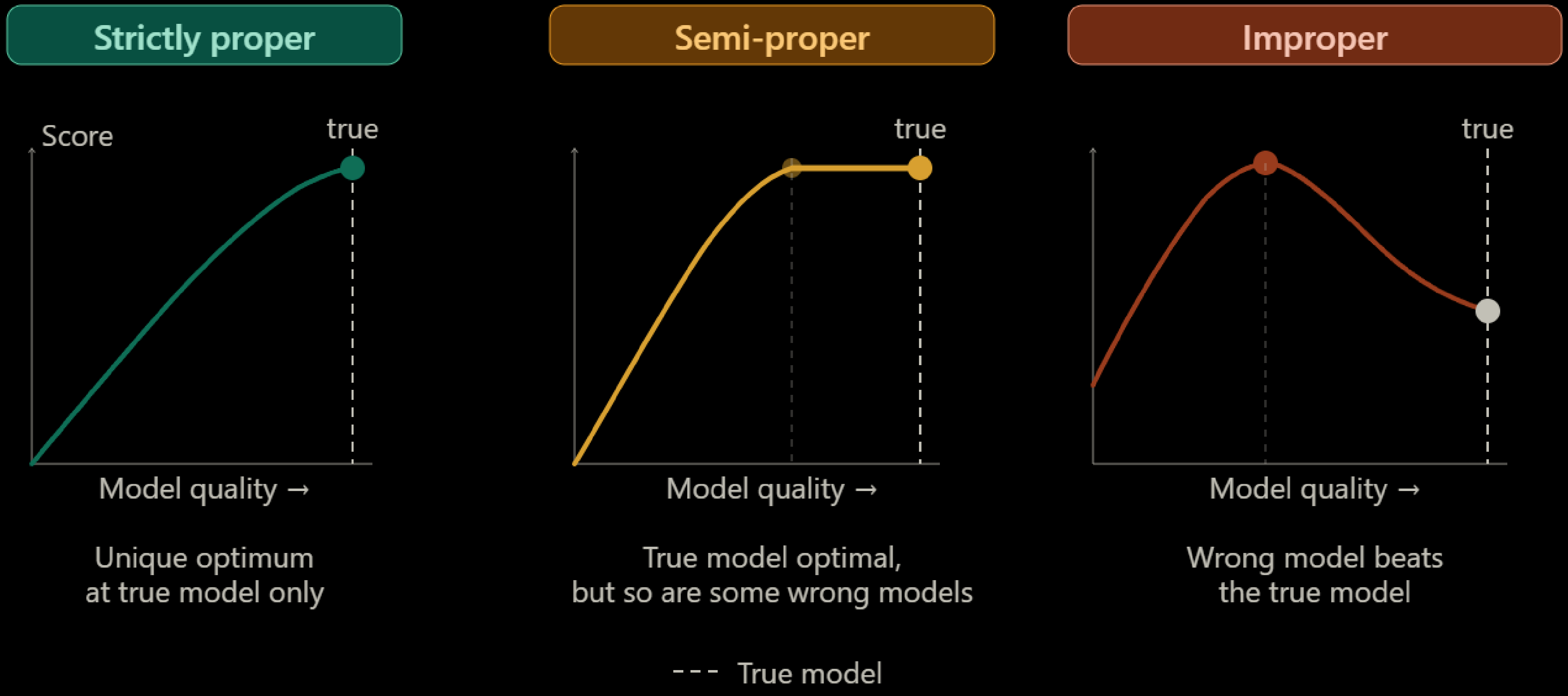
Properness

- A performance metric is proper if it is optimized on expectation by using ”the correct model”
- That is, it is maximized when $\hat{P}$ = P
- Of course, in any given iteration, random variability might lead to a “non-correct” modeling beating a correct one, but we are discussing expectations

Properness (2)

- We discuss three levels:
  - Strictly Proper – the expected value is optimal only for the correct model
  - Semi-proper – the expected value is optimal for the correct model and for some incorrect models
  - Improper – Some incorrect models have better expected value than the correct model
- This makes intuitive sense – what more can we hope from our model than to predict the true probabilities?
- A metric that is improper can be “fooled” or “gamed” – we can improve it on expectation with an incorrect model

Properness (2)



Clear Focus

- The predictive performance of models is evaluated in two regards
- First, we evaluate the statistical performance, disconnected from the eventual use of the model
- Second, we incorporate ”costs” from the intended use of the model, and evaluate the model’s utility for decision making
- Accordingly, metrics should either be purely statistical, or should explicitly incorporate the concept of costs from the domain
- A metric that is implicitly affected by other “costs” is neither here nor there, and fails in providing helpful information in both stages

Interpretability

- The vaguest of the desiderata, but of immense practical importance
- Basically – if I tell you the number, do you intuitively understand whether the model is good or bad, and what needs to be fixed
- Is heavily dependent on training, so we'll mostly be thinking of "interpretable to people with a background in medicine/epidemiology/biostatistics" (=your clients)

Discrimination

- The first family
- Does the model assign higher probabilities to individuals with the event than for those without?
- It is a rank measure that reflects “relative” performance
- That is, it doesn’t care how high or low the absolute predicted probabilities are, only whether they discriminate the ”events” from the “non-events”

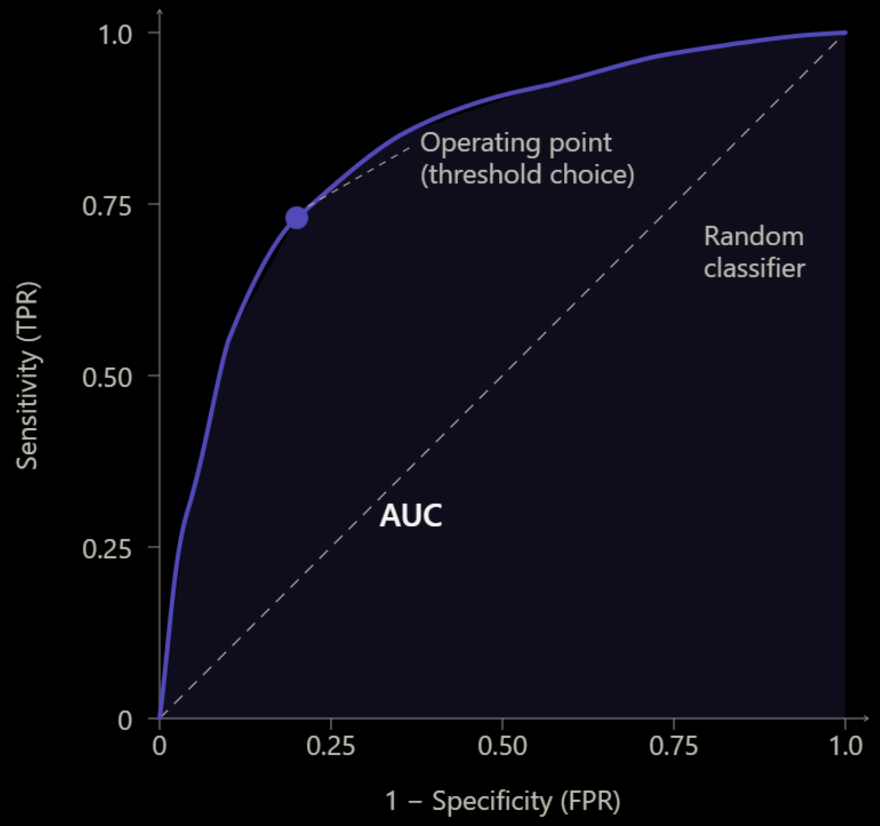
Discrimination - Justification

- Discrimination seems hard to justify, because it is only based on ranking (=is invariant under any monotonic transformation)
- I.e., it doesn't care if we take the root of all the predictions
- But quite often, in medicine, it's all we care about, why?
- Use case #1 - we believe our intervention is of benefit to everyone receiving it, but we have limited resources and cannot offer it to everyone
- And like we said, we choose those at maximum risk

Receiver Operating Characteristic (ROC) Curve

- Plotting the sensitivity against one minus the specificity for different cut-off values yields the ROC curve
- A diagonal line means random guessing
- The area under the ROC curve (AUROC, or just AUC) is… the area under the curve
- Integrating over the full range of thresholds, it is the quintessential discrimination measure, and often quoted as the sole performance measure of a given model
- 0.5 is random guessing, 1 is perfect discrimination

Receiver Operating Characteristic (ROC) Curve



AUC = 0.5
Random guessing

AUC = 1.0
Perfect discrimination

AUC = 0.70–0.80
Typical clinical model

P(score(yes) > score(no))
for a random positive–negative pair

AUROC – Alternate Interpretation

- For a binary outcome, the AUROC equals the ″c-statistic″, which has a nice probabilistic interpretation
- For a pair of observations, one “yes” and one “no”, what is the probability that the yes gets a higher prediction than the no?

AUROC – Alternate Interpretation (2)

X ~ Distribution of scores among positives, Y ~ Distribution of scores among negative, X ⫫ Y
Assume P(X=Y)=0 (no ties)

$$P(X > Y) = E_Y[P(X > Y|Y)] =$$ *[Law of Total Expectation]*

$$E_Y[P(X > t|Y = t)] = E_Y[P(X > t)]$$ *[Independence]*

$$\int_{-\infty}^{\infty} P(X > t) f_Y(t)\, dt =$$ *[Definition of Expectation]*

$$\int_{-\infty}^{\infty} \mathrm{TPR}(t)\, f_Y(t)\, dt =$$ *[TPR = P(X>t)]*

$$-\int_{-\infty}^{\infty} \mathrm{TPR}(t)\, d\, \mathrm{FPR}(t) =$$ *[$f_Y(t)\, dt = -d\, \mathrm{FPR}(t)$]*

$$\int_{0}^{1} TPR\ d(FPR) =$$ *[When t goes from minus infinity to infinity, FPR goes from 1 to 0]*

$$AUROC$$

Dependence on Baseline Prevalence

- AUROC is sometimes frowned upon because it is not sensitive to the baseline proportion (i.e., it conditions on “a positive and a negative”)
- More concretely, you can have a high AUROC, but still find it very hard to get reasonable recall and precision values if the outcome is rare (“class imbalance”)
- We do not consider this a problem. AUROC is a statistical measure that is not meant to take other costs into account

AUROC – Scorecard

- Properness? **Semi-Proper**
  - Has to be, since it’s a rank measure
- Clear Focus? **Focused**
  - Purely statistical
- Interpretability? **Yes**
  - Both for the intuition (e.g., people intuitively feel what an AUC of 0.52 means) and for the probabilistic interpretation
- Bottom Line? **Recommended**

Is AUROC's Kingship Justified?

- Mostly a resounding yes
- Why?
  - Good interpretability for the meaning and the actual number
  - Integrates over the entire range of thresholds
  - Does not depend on the baseline prevalence
  - A lot of decisions are of the first kind, meaning we only care about ranking
  - Discrimination is in some ways the heart of a prediction model, and AUROC captures it very nicely
- Bottom line – if I can only get one performance measure, AUROC is a great option

Other Discrimination Measures

- The area under the precision-recall curve (AUPRC) is sometimes positively presented as an "AUROC equivalent that is sensitive to baseline rates"
- This is not the prevailing position in clinical modeling. By incorporating the baseline proportion, AUPRC loses "Clear Focus" – it now mixes statistical performance with clinical utility, albeit without a consciously chosen threshold
- It also ignores true negatives

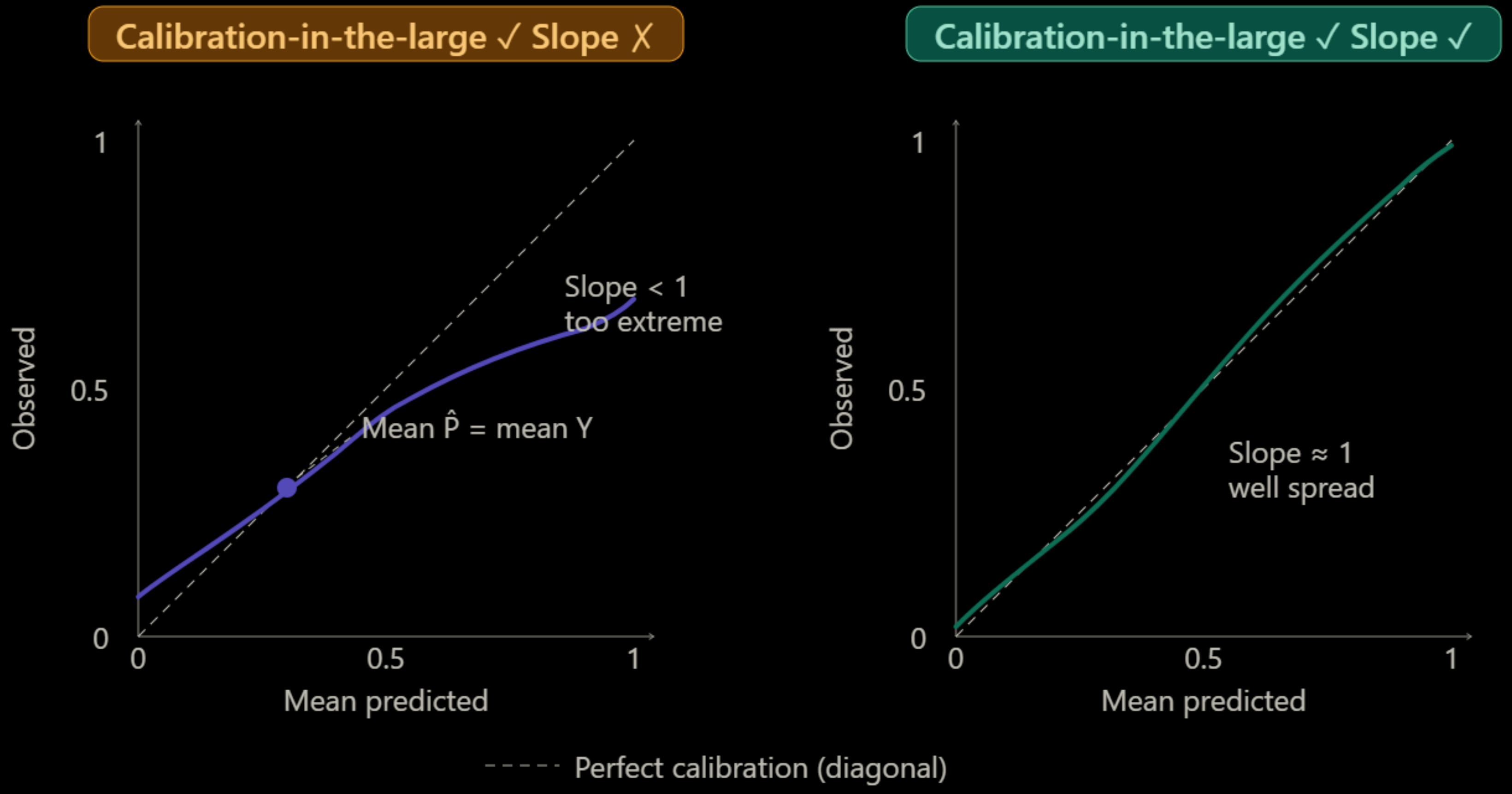
Calibration

- The second family
- Focusing on the latent P (and not the observed Y), does $\hat{P}$ = P?
- The obvious problem is of course that P is not observed, so how can we do this?

Simple Calibration Measures

- The simplest measure of calibration is to check that E[$\hat{P}$]=E[Y] across the entire sample, this is called **calibration-in-the-large**
- While useful, this can be satisfied trivially by predicting $\hat{P}$=E[Y] for everyone, which is nothing more than a null predictor
- Thus, an additional measure places the linear predictor from a logistic regression model (logit of predictions) as a sole predictor in a logistic regression model and checks for the slope (good = 1). This is the **calibration slope,** and it checks the spread of the predictions
- These are both rather "crude"

Simple Calibration Measures (2)

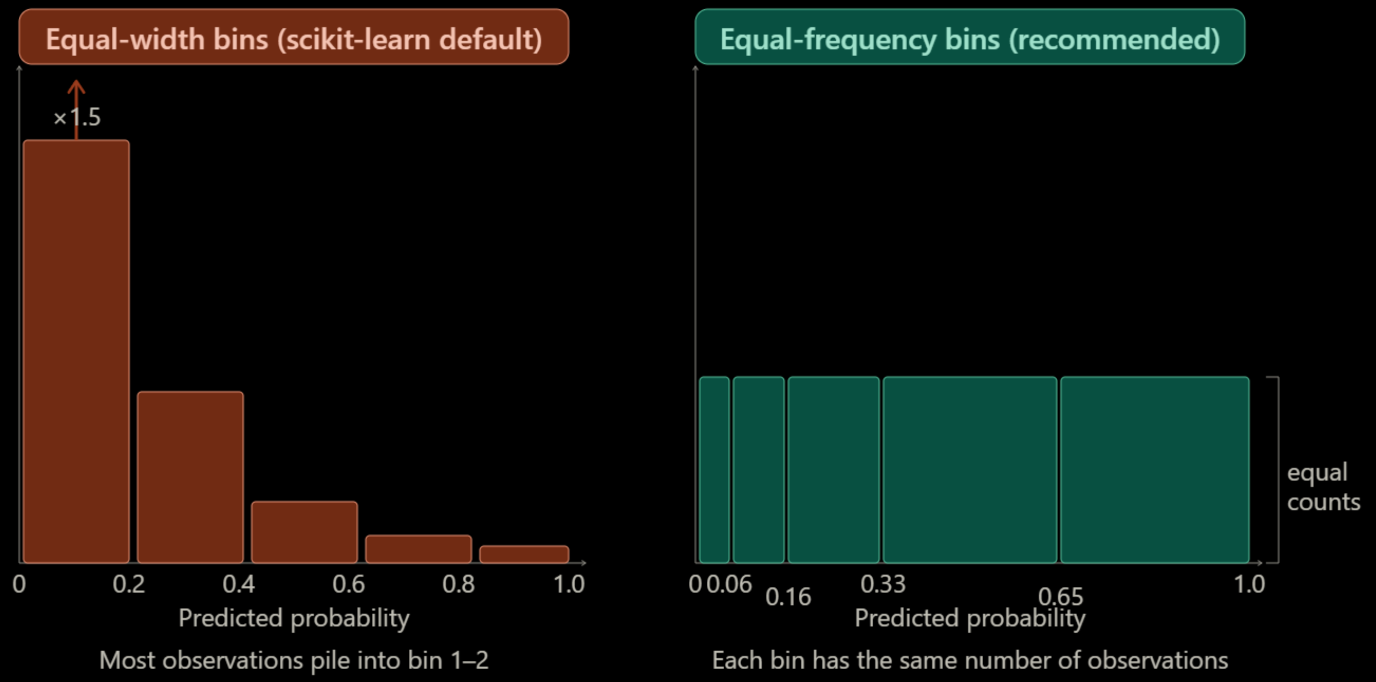


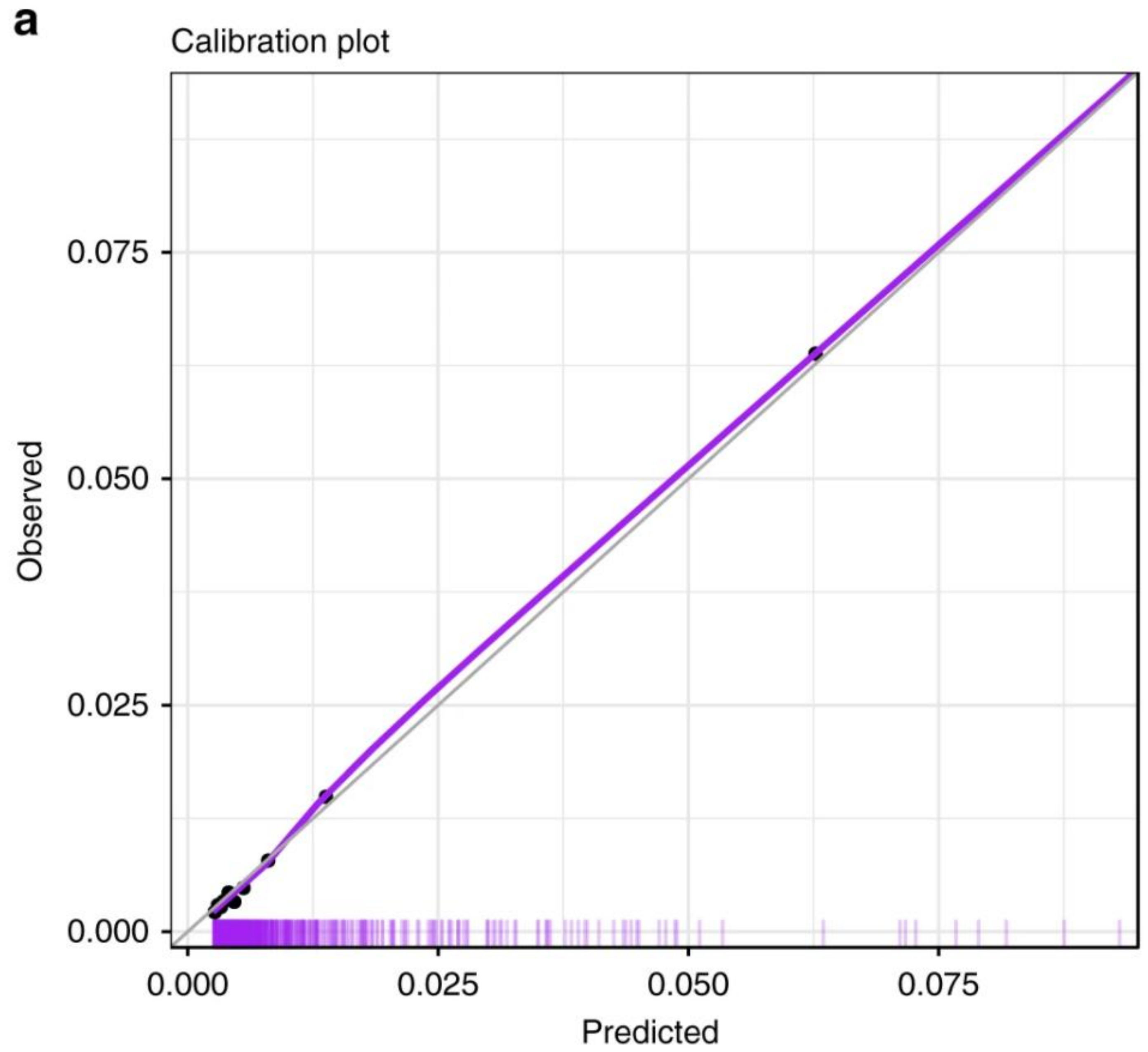
Calibration via Discretization

- So, we have Y, and we want P, how?
- The first way is to discretize. We take a group of patients with (somewhat) similar predictions, and check the outcome probability
- We expect E[$\hat{P}$]=E[Y] in each group
- The problem here is that discretization is inherently subjective. Do we use deciles? Quintiles? Something else? The result could change greatly

Calibration via Discretization (2)

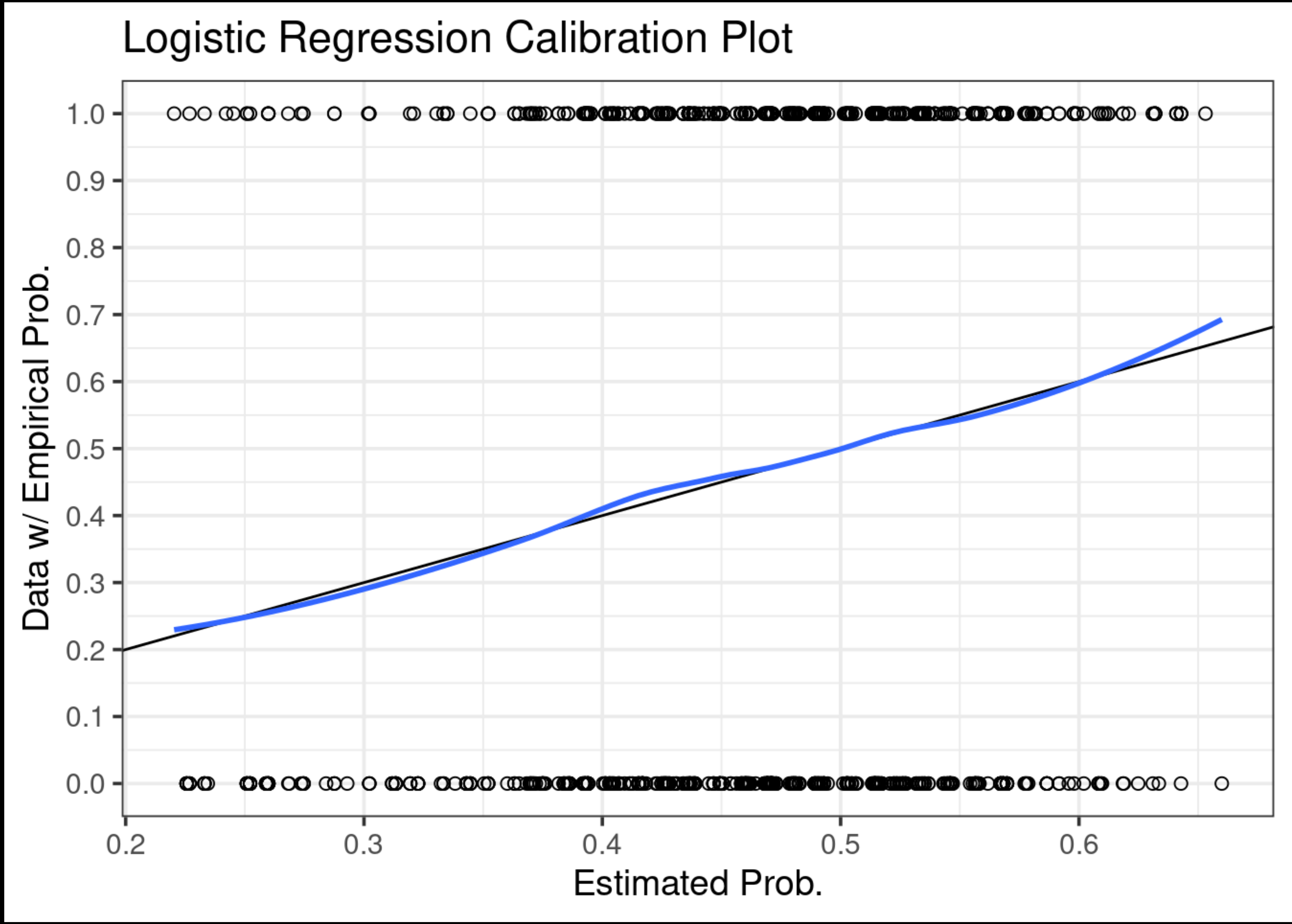
- If we do use this option (which we shouldn't), we should make sure to use the quantiles of the prediction to bin ("equal-frequency") and not absolute proportion cut-offs ("equal-width")




a
Calibration plot
Observed
Predicted
0.000
0.025
0.050
0.075
0.000
0.025
0.050
0.075

Smooth Calibration Plot

- In essence, the process going from Y => P is one of smoothing, the Y being a “coarse” version of P
- We can then use standard smoothing techniques, such as LOESS or splines
- And then compare this smoothed curve to the “optimal calibration line”, which is the diagonal
- This is usually done visually, but there are also scalar statistics


Logistic Regression Calibration Plot
Data w/ Empirical Prob.
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.2
0.3
0.4
0.5
0.6
Estimated Prob.

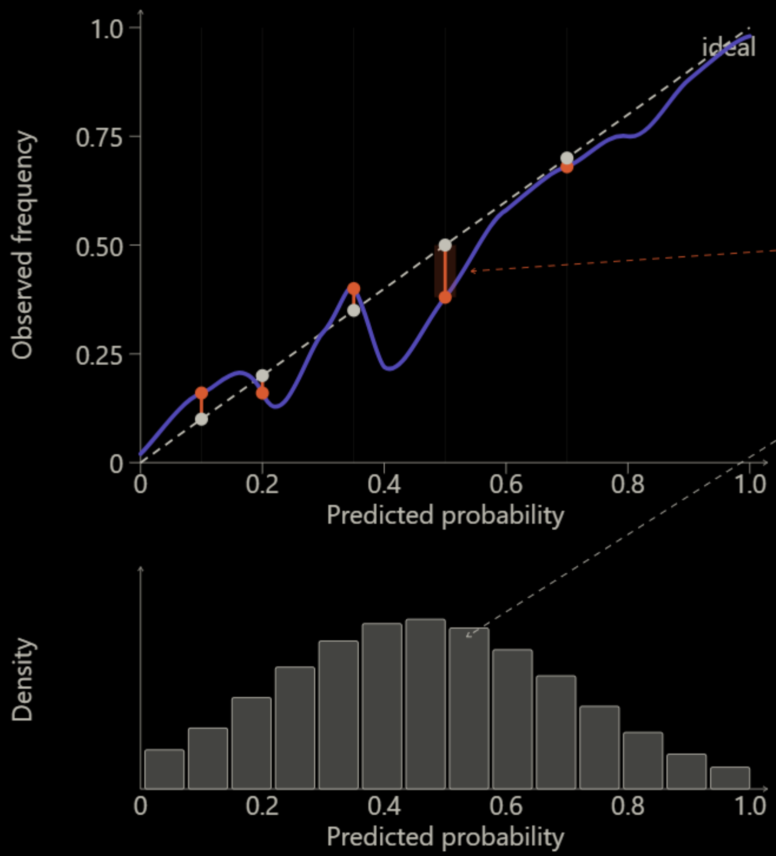
Smooth Calibration Plot - Caveat

- In fact, this is not really “objective”, because smoothing also has parameters
- For example, LOESS needs to know the bandwidth - how much should each point care about points that are further away?
- Splines have the number of knots and the degree

Calibration Summary Measures

- There are various measures aiming to quantify calibration in a single number
- Most famous is the Integrated Calibration Index (ICI), which is an integral of the distance between the calibration plot and the diagonal line, integrated over the empirical distribution of predictions
- Because these single-number-summaries do not tell us the direction of miscalibration, we generally prefer just showing a smooth calibration plot

Calibration Summary Measures



ICI = E[|Cal(P) − P|]

Average absolute distance, weighted by the empirical distribution of predictions

1. **Distance to diagonal**
   Measured vertically at each predicted p
2. **Weighted by prediction density**
   Common predictions contribute more
3. **ICI is small when the curve hugs the diagonal**
   Especially where predictions are dense

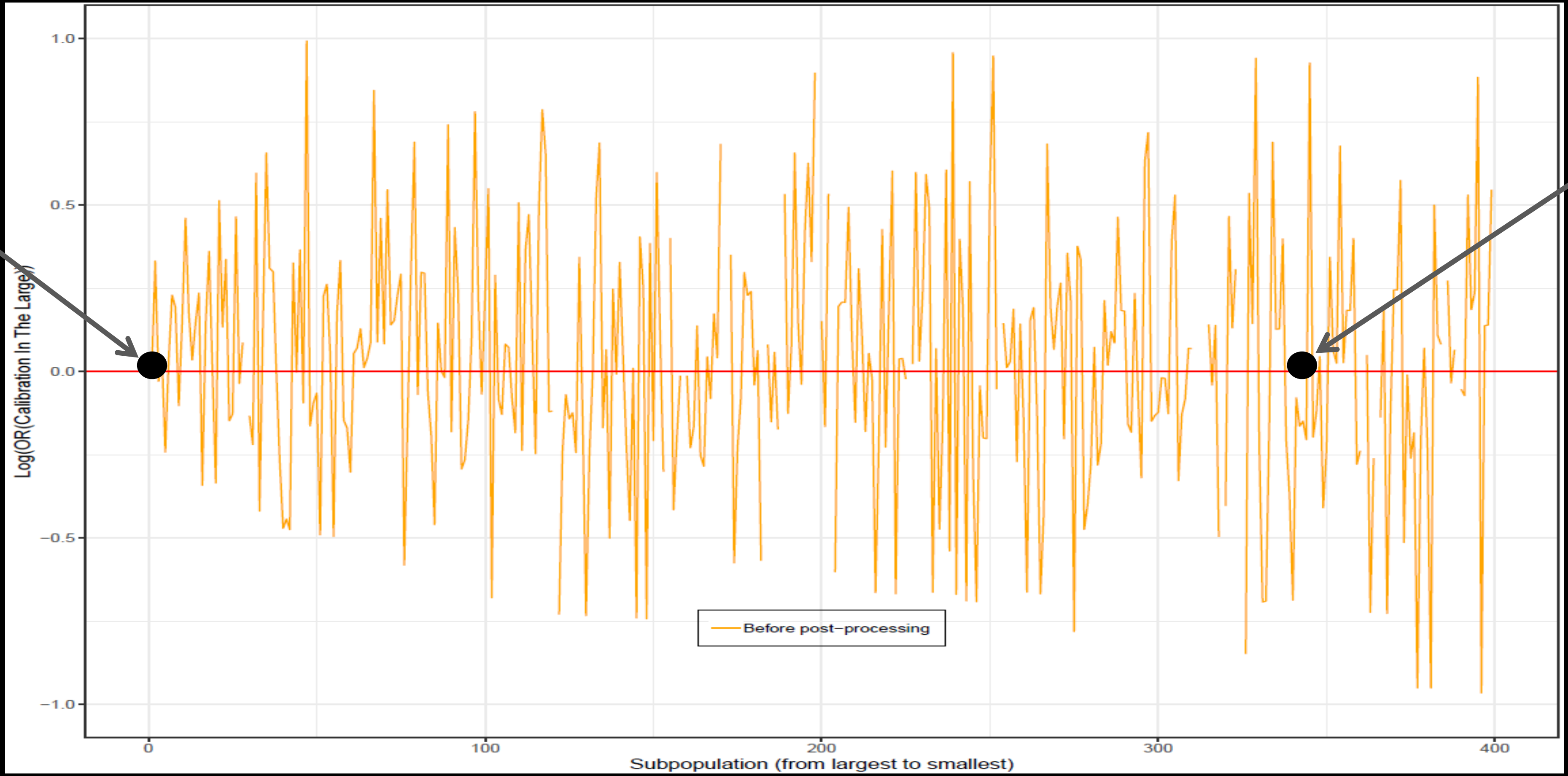
- - - Ideal calibration
— Estimated calibration curve
— Distance |Cal(p) − p|

Multicalibration (=Strong Calibration)

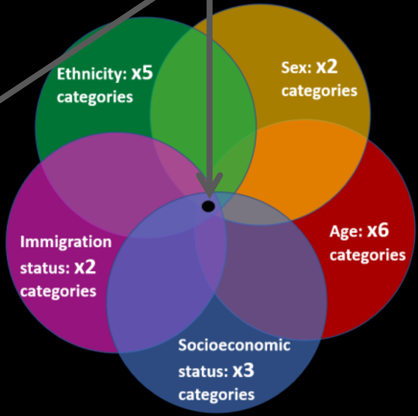
- Because calibration is never an individual-level metric, the choice of the reference population is very important
- For a given person, who do we average him with?
- Increasingly it is becoming evident that we need to check calibration separately for different subgroups, because models are often well-calibrated for the entire population (and maybe large subgroups), but not for specific subgroups, specifically minority subgroups
- We'll discuss further in the fairness lecture

Multicalibration (=Strong Calibration)

Entire population



A subpopulation of Caucasian females, 50-59 years of age, of low socioeconomic status and a history of immigration



Recalibration

- Many model families tend to output miscalibrated predictions
- But there is nothing stopping us from recalibrating as a post-processing step
- The simplest way is to use the predictions (or their logit) as the sole predictor in a logistic regression model, and then to “predict” the new prediction as the output from this second model
  - This is “linear recalibration” (“Platt calibration”) – only changes the slope and intercept
- There are other, fancier techniques
  - But usual caveats about overfitting still hold – the more expressive your models are, the more you are likely to overfit

Recalibration (2)

- Note that we can recalibrate models in post-processing, but we can’t “rediscriminate” them
- This is, perhaps, another reason for the popularity of AUROC

ICI – Scorecard

- Properness? **Semi-Proper**
  - Because it conditions on the predictions, a base-rate predictor will be perfectly calibrated
- Clear Focus? **Focused**
  - Purely statistical
- Interpretability? **Yes**
  - Less so for the ICI itself, but definitely for a smooth calibration plot
- Bottom Line? **You should always draw a smooth calibration plot**

Overall Performance ("Loss")

- The third family
- Most famous are the cross-entropy and Brier score
- But also includes things like R-squared (and its variations) and others

Log Loss / Cross-Entropy / Deviance / Log-Likelihood

- The log-likelihood. The indisputable mother of all performance metrics and what most models directly train on
- For our binary setting, defined as $-[y \log p + (1-y)\log(1-p)]$
  - More generally as $-\log q(y)$, the probability/density at the observed outcome
- Is strongly grounded in information theory (it is the KL divergence from the true model + the constant unknown information entropy of the baseline distribution)

Log Loss / Cross-Entropy / Deviance / Log-Likelihood

- In more formal probabilistic settings (e.g., Bayesian statistics), is often the central metric people care about
- However, it unboundedly punishes strong mistaken predictions (a probability of zero on the observed outcome gives an infinite loss)
- And is not directly interpretable (only in comparisons)

Brier Score

- Related to the binary cross-entropy: $\frac{1}{N}\sum_{i=1}^{N}\left(\hat{P} - Y\right)^2$
- Decomposes nicely to metrics of discrimination and calibration (Murphy's decomposition):
  - Brier = Reliability – Resolution + Uncertainty
  - Reliability = $\frac{1}{N}\sum_K n_k(p_k - \bar{y}_k)^2$ *[calibration error]*
  - Resolution = $\frac{1}{N}\sum_K N_k(\bar{y}_k - \bar{y})^2$ *[how true event rate differs from overall]*
  - Uncertainty = $\bar{y}(1 - \bar{y})$ *[variance of actual outcome]*
- Less punishing on extremes

Brier’s Score / Logloss – Scorecard

- Properness? **Strictly Proper**
  - For multiclass problems, logloss is also the only strictly proper measure that is local
- Clear Focus? **Focused**
  - Purely statistical
- Interpretability? **No**
  - The number only makes sense in comparisons
  - They mix calibration and discrimination
- Bottom Line? **For training and model selection**

Classification

- The fourth family
- We use a clinically relevant threshold, discretize, and read the metrics off the resulting confusion matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Accuracy

- The simplest and least useful
- Once discretized, we have $\hat{Y}$ and Y, which we can simply compare
- Accuracy = (TP+TN)/N
- Fails miserably if the outcome is not balanced, because e.g., for a rare disease we can just predict “no” all the time and be right 99% of the time
- Reporting accuracy for an unbalanced outcome is a strong sign of not knowing what one is doing – be wary

The Classic Classification Metrics

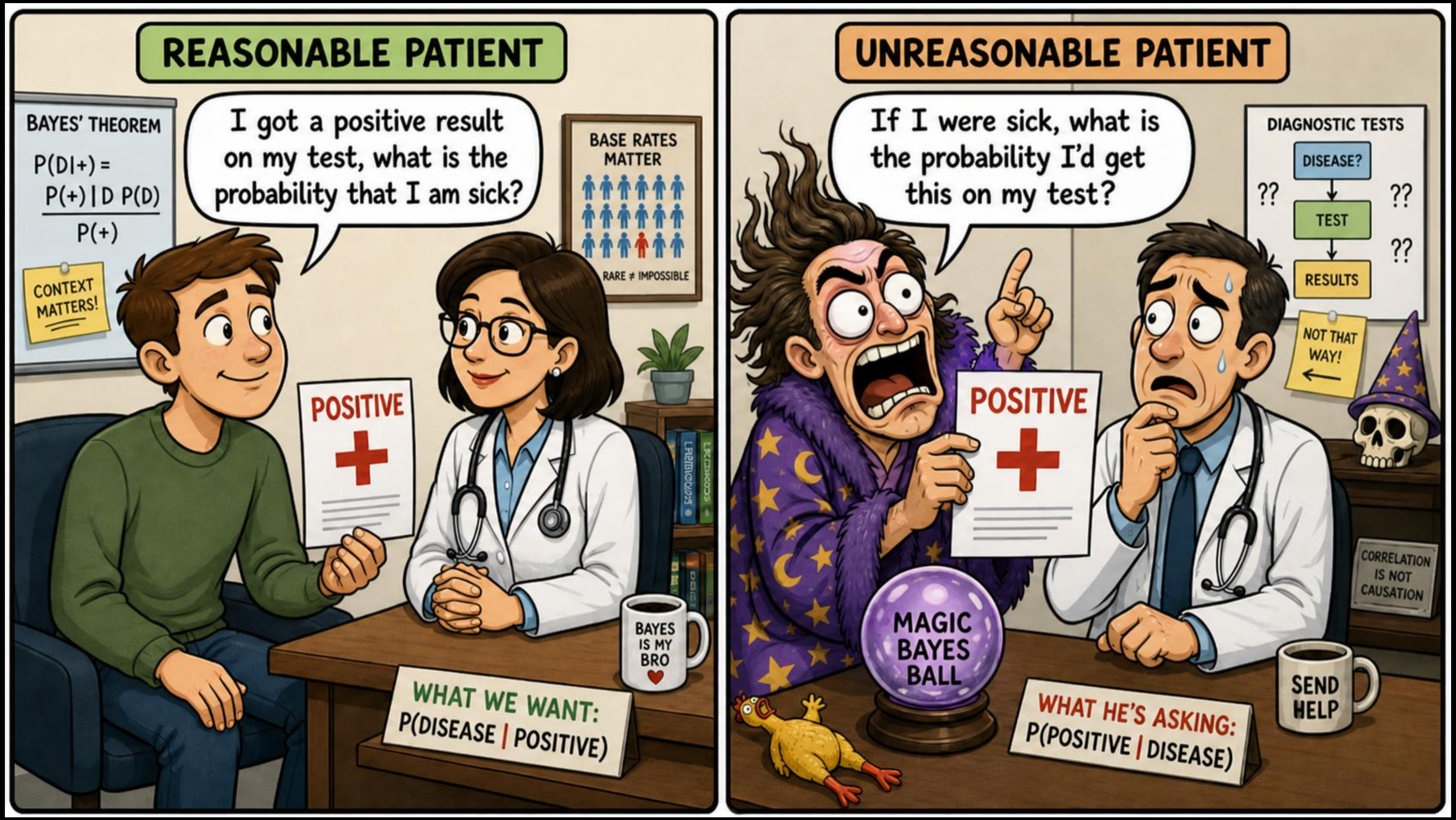
- The most common measures of classification are:

1. Sensitivity (Recall/TPR) – of those that experienced the outcome, what proportion did I identify?
2. Specificity (TNR) – of those that did not experience the outcome, what proportion did I identify?
3. Positive Predictive Value (PPV; Precision) – of those which for I predicted “yes”, what proportion experienced the outcome?
4. Negative Predictive Value (NPV) – of those which for I predicted “no”, what proportion did not experience the outcome?

Sens/Spec vs. PPV/NPV

- Sensitivity and specificity have the advantage of being invariant to the baseline rate (that is, they are only a function of the test being used)
- If I now relocate to a place where the baseline rate is 1/10, they don't change
- They have the disadvantage of being conditioned on the true state, which I don't know, which makes them somewhat useless in real life
- PPV/NPV, on the other hand, are very dependent on the baseline rate, but are conditioned on the test result, which I know, which makes them very useful in actual use

Sens/Spec vs. PPV/NPV



Sens/Spec vs. PPV/NPV - Example

Baseline prevalence = 10% — 100 events in 1000 patients

	Actually positive	Actually negative	
Predicted positive	50 True positives	500 False positives	PPV = 9% 50 / 550
Predicted negative	50 False negatives	400 True negatives	
	Sensitivity = 50% 50 / 100	Specificity = 44% 400 / 900	

Even with sensitivity 50%, most positive predictions are wrong when the disease is rare.

Sens/Spec vs. PPV/NPV - Example

Baseline prevalence = 30% — 300 events in 1000 patients

	Actually positive	Actually negative	
Predicted positive	150 True positives	350 False positives	PPV = 30% 150 / 500
Predicted negative	150 False negatives	350 True negatives	
	Sensitivity = 50% 150 / 300	Specificity = 50% 350 / 700	

Same test, same sensitivity — PPV triples as prevalence rises from 10% to 30%.

Sens/Spec vs. PPV/NPV - Example

Baseline prevalence = 50% — 500 events in 1000 patients

	Actually positive	Actually negative	
Predicted positive	**250** True positives	**250** False positives	**PPV = 50%** 250 / 500
Predicted negative	**250** False negatives	**250** True negatives	
	Sensitivity = 50% 250 / 500	**Specificity = 50%** 250 / 500	

Sensitivity and specificity unchanged — PPV went from 9% → 30% → 50% purely from prevalence.

Trade-offs

- It is very intuitive to understand that there is a trade-off between sensitivity and specificity, as we set the threshold higher or lower
- If we just say that everyone is “yes”, our sensitivity will be 100% (we won’t miss anyone), but our specificity would be 0% (we would err on all the “no”s)
- Same thing the other way
- Also note that for relatively rare diseases (which is most diseases), PPV is most affected by specificity

Sens/Spec/PPV/NPV – Scorecard

- Properness? **Improper on their own**
  - Thus, must never be shown in isolation
- Clear Focus? **Focused**
  - Purely statistical
- Interpretability? **Yes**
  - Clinicians, specifically, are very accustomed to these metrics
- Bottom Line? **Always in groups, and always when anchored to a clinically relevant threshold**

Sensitivity/PPV/Predicted Positive Curves

- When deciding if performance at a given threshold is acceptable, a useful triptych to draw is one that plots Sensitivity, PPV and predicted positive one against the other
- This allows us to say things along the lines of "for a given threshold"
  - We mark 50% of the population as being "at risk"
  - We successfully catch 10% of the cases
  - For each case we catch, we mistakenly identify 2 false positives
- This is usually the language that is easiest for domain experts (e.g., clinicians) to understand

A (unrelated) Triptych

A One Plot Triptych

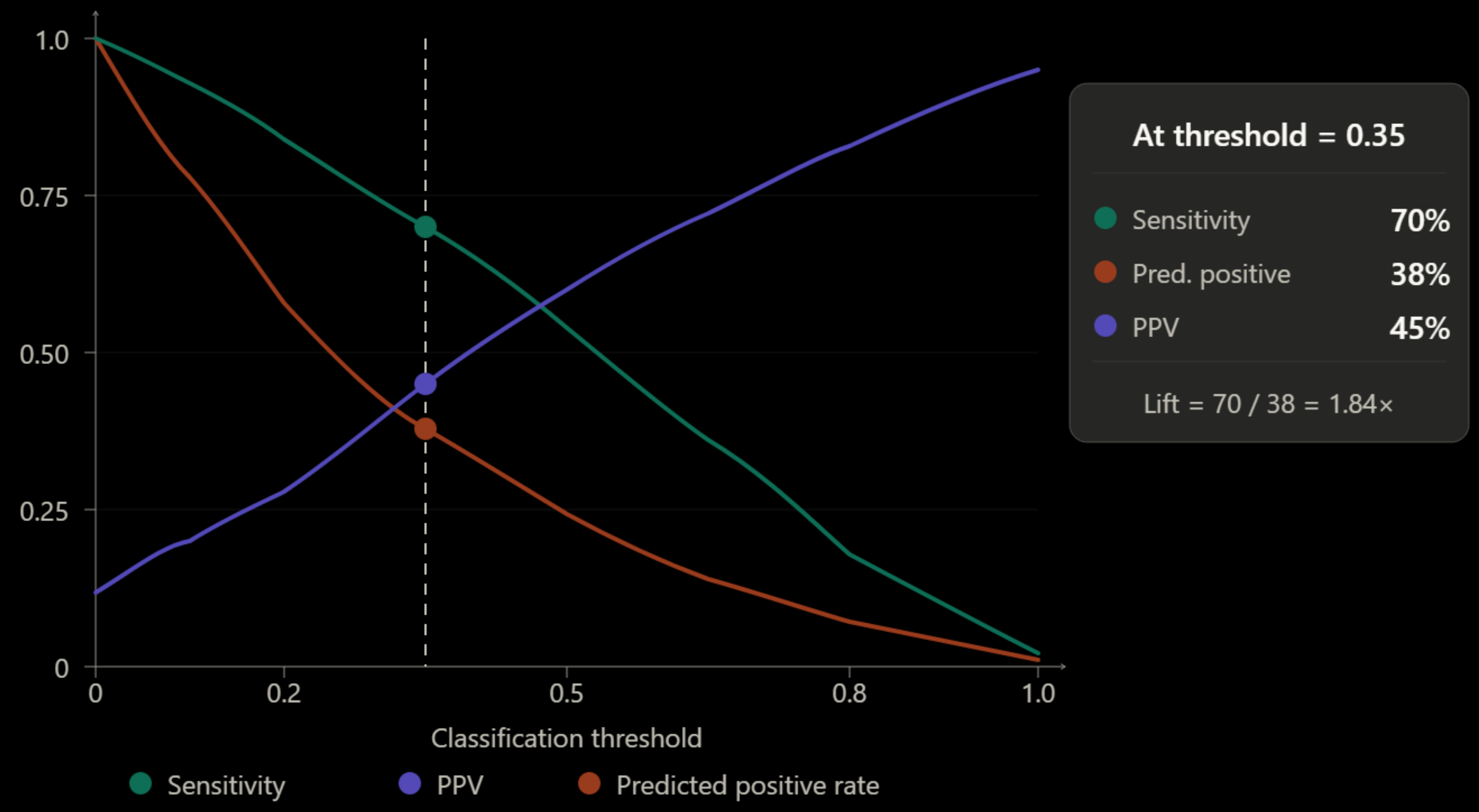
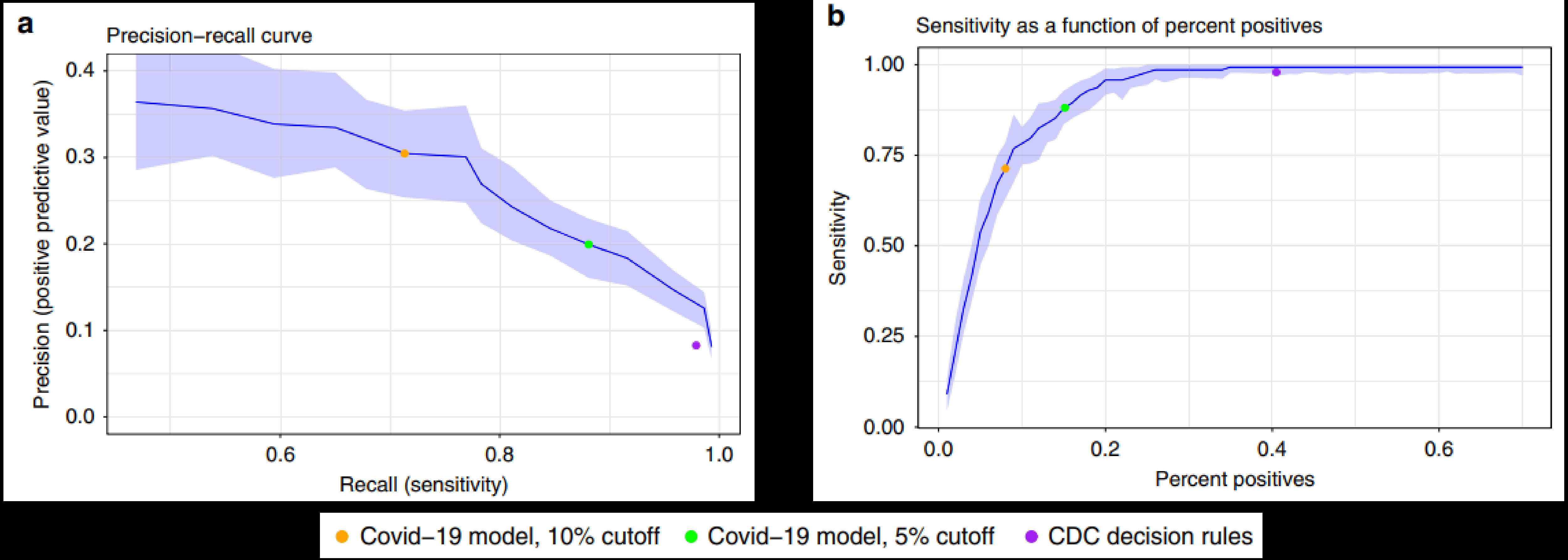
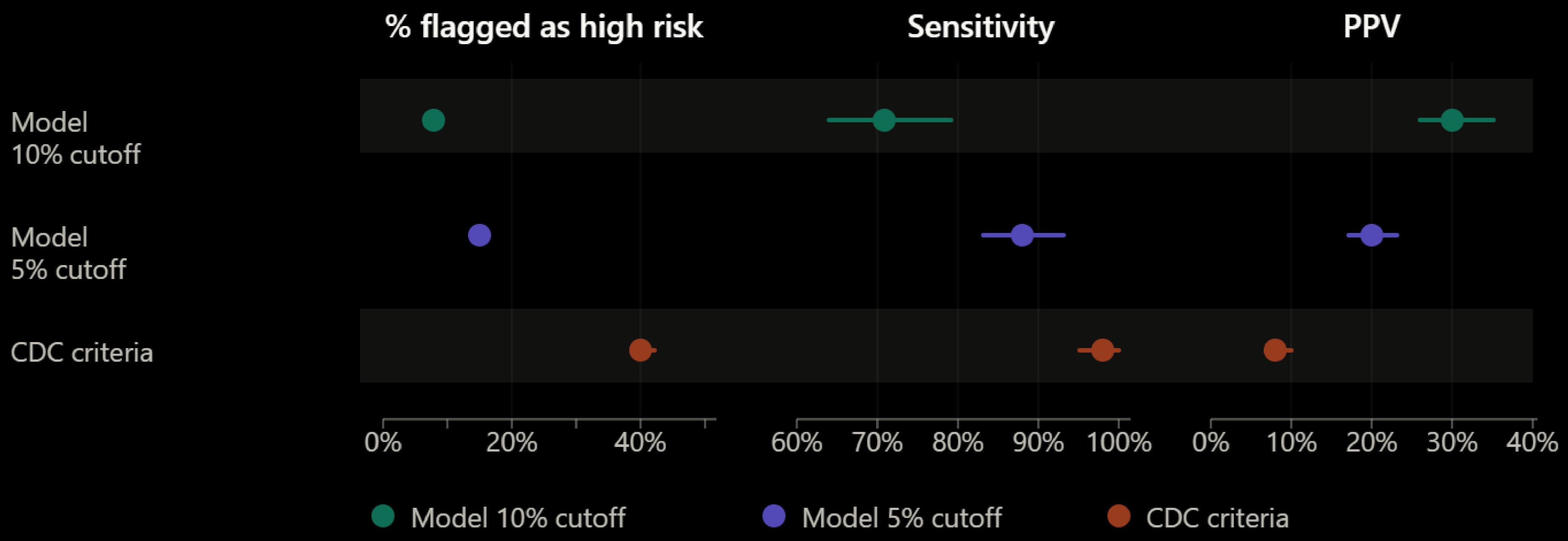


Fig. 3 Performance charts for the COVID-19 model.



Noam Barda, Dan Riesel, Amichay Akriv, Joseph Levy, Uriah Finkel, Gal Yona, Daniel Greenfeld, Shimon Sheiba, Jonathan Somer, Eitan Bachmat, Guy N. Rothblum, Uri Shalit, Doron Netzer, Ran Balicer & Noa Dagan. Developing a COVID-19 mortality risk prediction model when individual-level data are not available. *Nat Commun* **11,** 4439 (2020). https://doi.org/10.1038/s41467-020-18297-9

Real-World Example

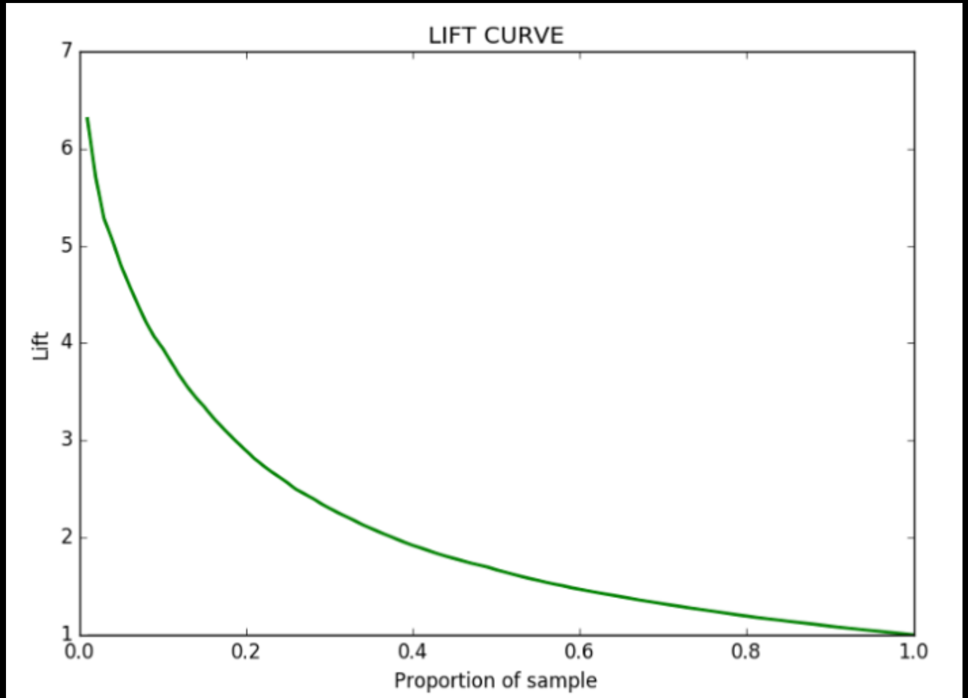


Lift

- If our intention is to “enrich” the population with positives, we can measure that directly
- Lift = $\frac{PPV}{Prevalence}$
  the ratio between the proportion of true positives from the selected population to the proportion of true positives from the general population
- Lift = $\frac{Sensitivity}{Percent\ Positive}$
  the ratio between the proportion of true positives we manage to identify to the proportion we tagged as high risk
- For any reasonable model, lift should increase as we increase the threshold

Lift (2)

- Lift is interpretable – a lift of 4 for example means we can “enrich” the population with positives 4 times over the general population
- Lift curves are pretty common, but remember that lift is just PPV scaled by the baseline prevalence



Reclassification

- Comparing two prediction models (or a model and some "null" classifier), if we mostly care about the binary classification, we can focus on that
- A "case" reclassified as positive gets +1, reclassified as negative gets -1
- A "control" reclassified as negative gets +1, reclassified as positive gets -1
- We average the results in each group (events and non-events), then sum the two numbers together
- This is the (Categorical) Net Reclassification Improvement (NRI)

Reclassification (2)

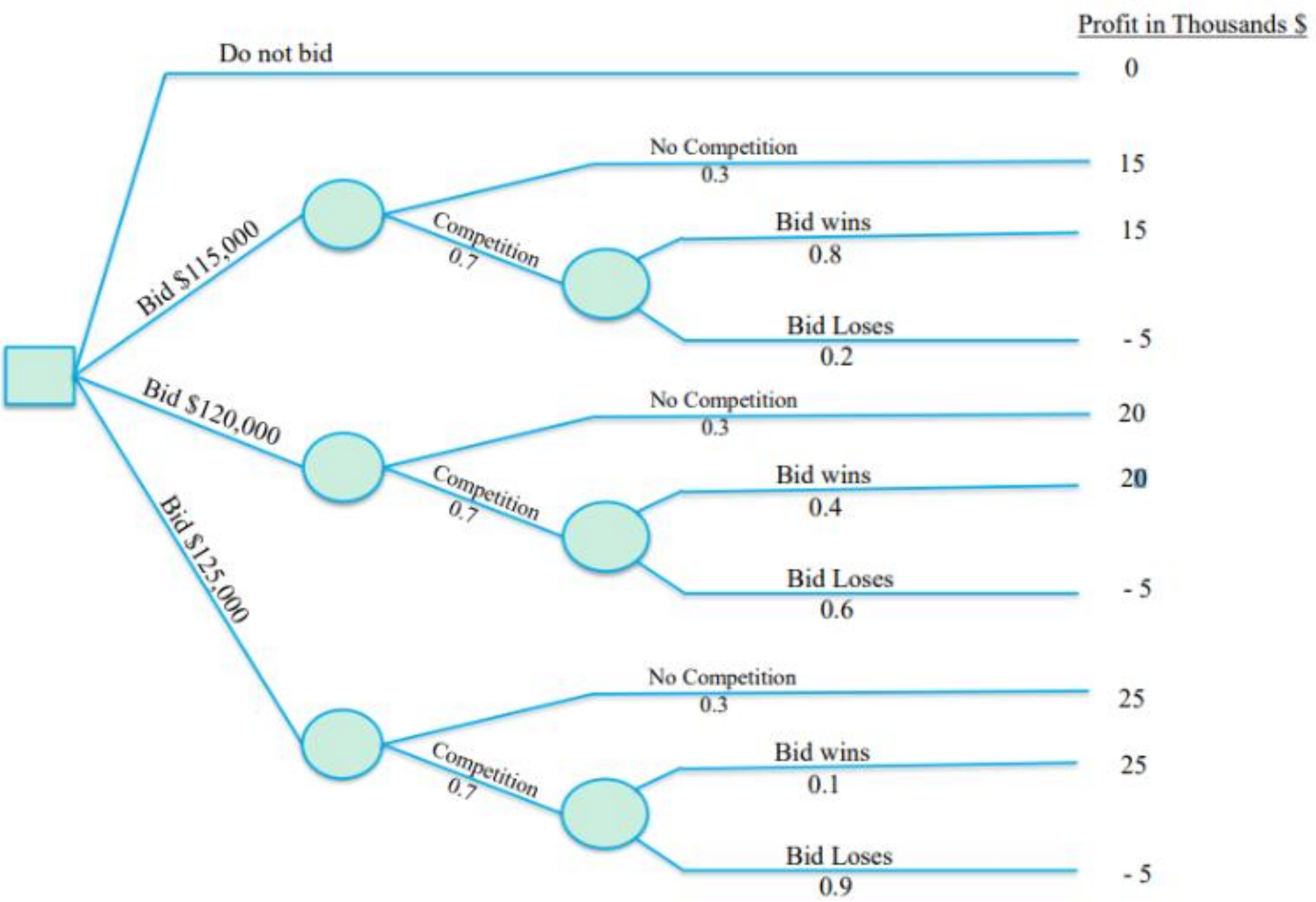
- Categorical NRI has come under fire because
  - It is sensitive to the base rate
  - It is sensitive to the number of categories
  - Empirical studies that show that it can be improved even by the addition of "null predictors" (predictors that are in no way associated with the outcome)
- This is of course a bad thing. It makes the metric very subjective and means that it can "be gamed"

Reclassification (3)

- And so was born Continuous NRI (cNRI), where any movement up for an event is good and any movement down is bad (and the opposite for non-events)
- Are we better off?
  - Good: Not gameable
  - Bad: epsilon up for an event is counted the same as +90%
- But this is what cNRI is, and it does enjoy popularity in clinical journals

Clinical Utility

- The fifth family
- Think decision science - our aim is not to predict, our aim is to make good decisions
- Under expected utility theory (EUT), the optimal decision is the one where the product of the probability of that outcome and the utility we allocate to the outcome is the maximal
- This assignment of utilities happens “before” any model. The model only estimates the probabilities



Profit in Thousands $
Do not bid
0
Bid $115,000
No Competition
0.3
15
Competition
0.7
Bid wins
0.8
15
Bid Loses
0.2
- 5
Bid $120,000
No Competition
0.3
20
Competition
0.7
Bid wins
0.4
20
Bid Loses
0.6
- 5
Bid $125,000
No Competition
0.3
25
Competition
0.7
Bid wins
0.1
25
Bid Loses
0.9
- 5

Net Benefit

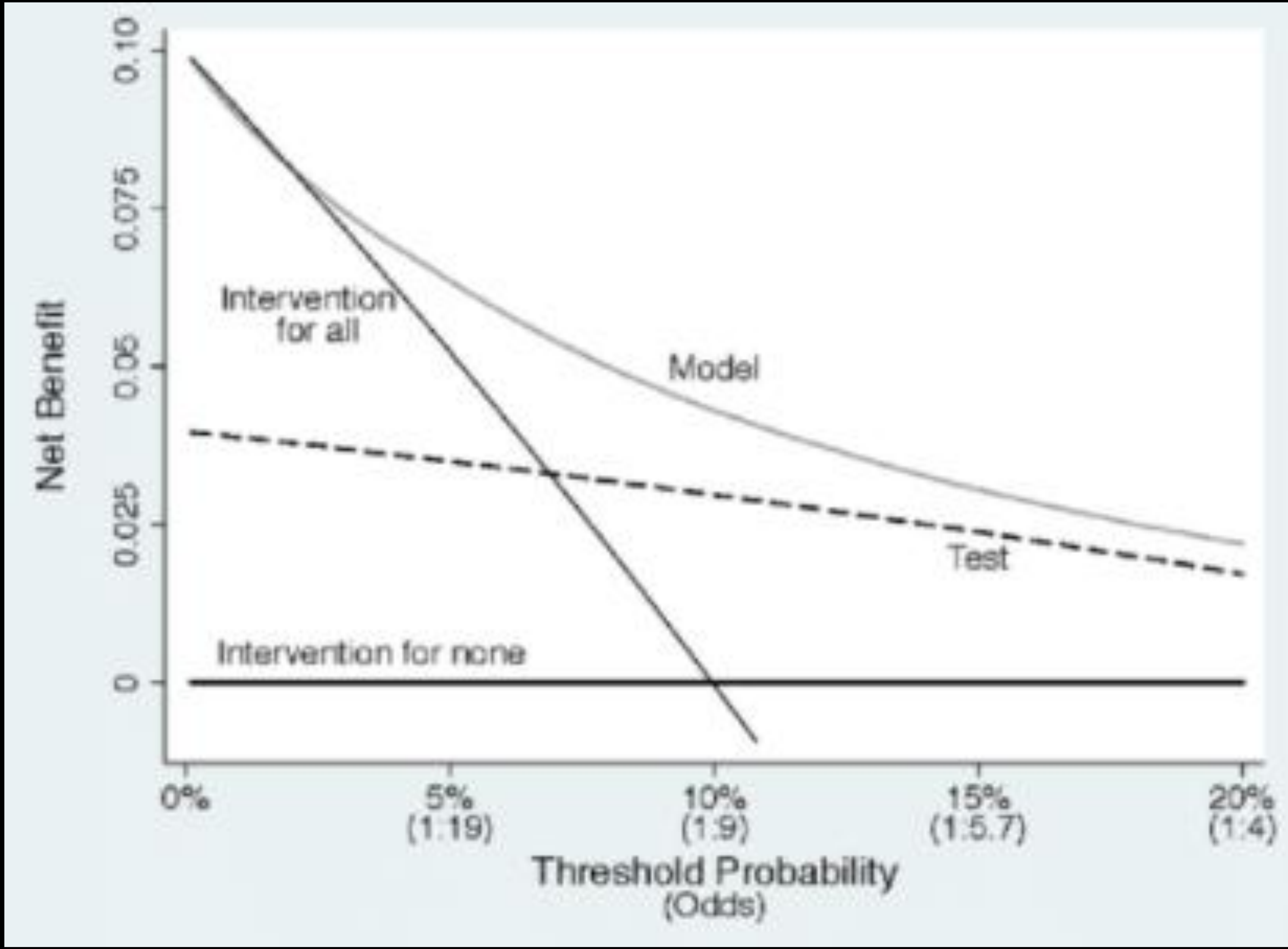
- Decision-theoretic analyses can be quite complicated, but in clinical prediction models an approximation called net benefit is sometimes used
- Logically, if I am willing to accept a probability threshold of 10%, I am willing to accept 9 false positives for each true positive
- We can then "read" the proportion of TPs and FPs at a given threshold and calculate the benefit of the model at that threshold

$$\text{Net benefit} = \frac{\text{True positives}}{N} - \frac{\text{False positives}}{N} \times \frac{p_t}{1-p_t}$$

- It has to be understood that given this approach, the threshold comes before the model
  - It is a function of the utilities we attribute to the different outcomes
  - It makes no sense to choose the point with the highest net benefit!

Decision Curve Analysis (DCA)

- We obtain a decision curve by plotting the net benefit for each model at each threshold
- Note that the threshold determines both the utility and the proportions of TPs and FPs
- We add two additional lines – treat no one and treat everyone
  - Treat no one has no positives, so the net benefit is always at 0
  - Treat everyone sees everyone as positive, so the net benefit is monotonically decreasing, as the price of the (constant number of) false positives continually rises



0.10
0.075
0.05
0.025
0
Net Benefit
Intervention for all
Model
Test
Intervention for none
0%
5% (1:19)
10% (1:9)
15% (1:5.7)
20% (1:4)
Threshold Probability
(Odds)

Net Benefit – Scorecard

- Properness? **Semi-proper**
  - Has to be, because of discretization
- Clear Focus? **Focused**
  - Purely decision-making
- Interpretability? **Yes**
  - Less so for raw net benefit, but good for the curve
- Bottom Line? **Recommended**

Scorecards Summary

Metric family	Properness	Clear focus	Interpretable	Verdict
Discrimination AUROC, c-statistic	Semi-proper	Focused	Yes	Recommended
Calibration ICI, smooth cal. plot	Semi-proper	Focused	Yes	Always plot
Overall Brier score, log loss	Strictly proper	Focused	No	Training / selection
Classification Sens, Spec, PPV, NPV	Improper alone	Focused	Yes	Always in groups
Clinical utility Net benefit, DCA	Semi-proper	Focused	Yes	Recommended

Positive Partial / conditional Negative

Uncertainty

- We have an entire presentation on that, so very generally
- Regardless of which measure you are reporting, these are just point estimates of parameters
- We need standard errors, confidence intervals, credible intervals, etc.
- Some of these statistics have theoretical distributions (almost always asymptotic), but a lot don’t, so we resort to techniques like bootstrapping
- For now, just consider that reporting point estimates is not enough

Interventions = Causal!

- Regardless of how good the model is, no metric of predictive performance can prove that the model would be helpful in real life
- Using a model in practice is an intervention, a policy, which means it requires causal inference, preferably an RCT
- Unless the benefit is very evident, you should become accustomed to answering a claim such as
  - "I have a very good model, can we deploy it in practice?"

  with
  - "has it been shown to be helpful in a well-conducted study?"

RESEARCH SUMMARY

Effect of Colonoscopy Screening on Risks of Colorectal Cancer and Related Death

Bretthauer M et al. DOI: 10.1056/NEJMoa2208375

CLINICAL PROBLEM

Although colonoscopy is widely used as a screening test to detect colorectal cancer, it is more invasive and requires more resources than fecal occult blood tests and sigmoidoscopy. To determine whether its benefits outweigh these costs, additional high-quality data are needed from randomized trials.

CLINICAL TRIAL

Design: A multinational, pragmatic, randomized trial assessed the 10-year effects of population-based colonoscopy screening on the risks of colorectal cancer and related death.

Intervention: 84,585 men and women 55 to 64 years of age who lived in Poland, Norway, and Sweden and who had not previously undergone screening were randomly assigned either to receive an invitation to undergo one-time colonoscopy screening (invited group) or to receive no invitation or screening (usual-care group). The primary end points were the risks of colorectal cancer and related death after median follow-ups of 10 years and 15 years.

RESULTS

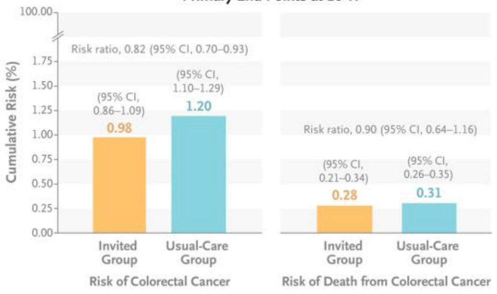
As compared with usual care, invitation to one-time screening colonoscopy reduced the risk of colorectal cancer over 10 years; 455 persons needed to be invited to undergo screening to prevent one case of colorectal cancer. Between-group differences in the risk of death from colorectal cancer were not significant.

LIMITATIONS

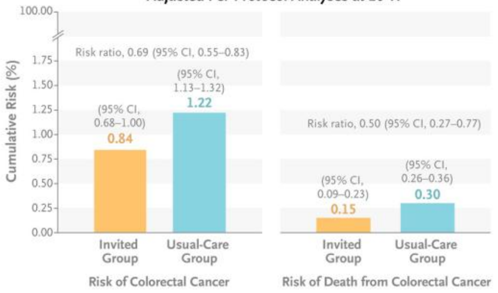
- Participation in screening was lower than expected in some countries (range, 33.0% to 60.7%).
- Information on adherence to recommendations regarding surveillance for polyps was lacking.
- Event rates were too low to assess variability according to quality indicators among endoscopists.
- Benefits with respect to the risk of colorectal cancer are expected to be apparent earlier than those with respect to the risk of death related to this disease; planned analyses after a median of 15 years of follow-up may be more informative in assessing the risk of death.

Links: Full Article | NEJM Quick Take | Editorial

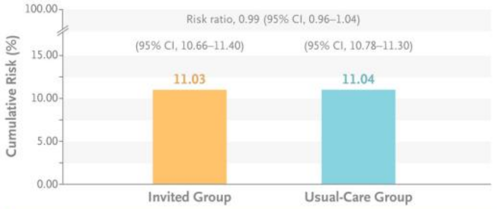
Primary End Points at 10 Yr



Adjusted Per-Protocol Analyses at 10 Yr



Death from Any Cause at 10 Yr



CONCLUSIONS

In participants 55 to 64 years of age who had not previously undergone screening, invitation to one-time screening with colonoscopy reduced the risk of colorectal cancer over a 10-year period.

Interventions = Causal! (2)

- You can’t get a new drug approved without formal causal studies (specifically RCTs)
- Despite this, for “ARAD” (אמצעים רפואיים דיגיטליים), when the risk of harm is considered low (i.e., the model merely shows a risk or reminds the clinician of an accepted treatment) there is currently no similar requirement
- Now-a-days, this is being discussed by the Israeli MOH with the help of a committee of domain experts